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I.6.6 Simulation Output Analysis

Files for peer review

All files submitted by the author for peer review are listed below. Files that could not be converted to PDF are indicated; reviewers are able to access them online.

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trust-by-construction-paper.pdf	Main Document - PDF	340.1 KB	Page 4

Trust by Construction: A Simulation Contract for Reproducible Computational Experiments

Joseph Krzywoszyja

***Abstract*—Computational experiments typically depend on export pipelines that separate solver and renderer into distinct tools with incompatible object models, creating an unformalized trust gap: what a researcher visualizes may silently diverge from what the solver computed. We propose *embodied simulation*, an architecture where render and solver share a single, hash-verified object model. We formalize this as a *simulation contract* with six enforceable guarantees: geometry integrity, unit validation, deterministic stepping, interaction provenance, auto-provenance, and exact replay. Trait composition and trust propagation follow a provenance semiring algebra with tropical strategies for conflict resolution, establishing a formal connection between simulation trust, graph optimization, and neural computation. We implement the contract atop a browser-native finite element framework with WebGPU compute shaders, quadratic tetrahedral elements (TET10), and built-in verification infrastructure aligned with ASME V&V standards. We demonstrate that contracted simulations satisfy the recently proposed CURE principles for computational models (Credible, Understandable, Reproducible, Extensible) and produce executable provenance records suitable as reproducibility packages for journal submissions. Our evaluation shows that contract enforcement adds less than 2% overhead to solver execution, superconvergent patch recovery (SPR) reproduces the NAFEMS LE1 benchmark stress to within 1.5%, and the unified architecture eliminates all format-conversion information loss inherent in traditional export pipelines.**

***Index Terms*—Scientific visualization, simulation reproducibility, finite element analysis, WebGPU, verification and validation, design by contract.**

I. Introduction

The computational sciences face a reproducibility crisis. A 2025 study found that fraudulent scientific publications are outpacing the growth rate of legitimate ones [1], while a peer-reviewed critique in *Communications of the ACM* exposed selective benchmarking and undisclosed tooling in a widely cited simulation study [2]. At the root of many such failures lies an architectural pattern so ubiquitous that it has escaped formal scrutiny: the *export pipeline*.

In the standard computational experiment workflow, a researcher defines geometry in a CAD tool, meshes it in a preprocessor, solves in a numerical engine, exports results to an interchange format, and visualizes in a post-processor. This pipeline introduces at least four format conversions, each of which may silently alter the data: coordinate systems may be reoriented, floating-point values truncated, mesh node orderings permuted by partitioning, physical units stripped, and temporal resolution reduced to the export sampling interval. Community reports confirm these failure modes are routine:

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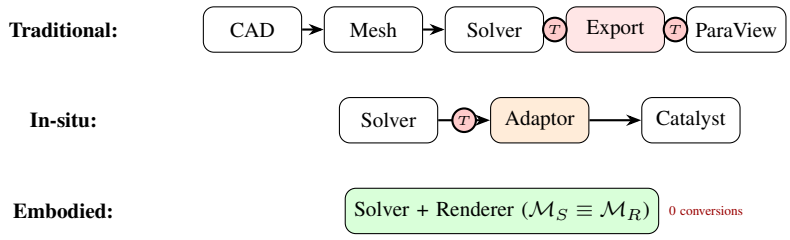

Fig. 1. Three simulation visualization architectures. Red circles mark format conversion boundaries (T). Traditional pipelines have 4+ conversions; in-situ reduces to 1 (the data adaptor); embodied eliminates all conversions by sharing the object model.

ANSYS result files are unreadable by ParaView without manual format editing; mesh point orderings diverge after domain decomposition; transient exports lose timestep metadata [3].

The fundamental issue is not one of software quality but of *architectural fidelity*. When the object model that the solver reads forces from is a different data structure than the object model the renderer reads vertices from, no amount of tool-chain discipline can guarantee that the visualization faithfully represents the computation. While the disconnect between solver and renderer state has been noted in the in-situ literature [5], [6], no prior work has formalized it as a contract or proposed architectural enforcement. We term this the *trust gap*.

We observe three levels at which this gap can be addressed:

1) **Trust by discipline.** The traditional approach: the researcher manually ensures compatible tool versions, export settings, and coordinate conventions. Trust depends on human diligence.
2) **Trust by co-location.** In-situ visualization frameworks [5], [6] embed the renderer within the solver process, eliminating file I/O. However, the solver and renderer still maintain separate object models connected by data adaptors. The adaptor boundary remains a conversion step, and no geometry identity is enforced.
3) **Trust by construction.** The approach we propose: the render model and the solver model are the *same semantic object*. Not exported copies. Not synchronized mirrors. The identical data structure that the solver reads displacement values from is the structure the renderer reads vertex positions from, verified at every step by hash verification.

Figure 1 contrasts these three approaches visually.

We formalize this architecture as a *simulation contract*—a runtime wrapper that enforces six guarantees on any numerical solver:

1) **Geometry integrity**: the solver mesh hash and render mesh hash are identical at every timestep.
2) **Unit validation**: all physical quantities carry validated units, range-checked against known material properties.
3) **Deterministic stepping**: a fixed-timestep accumulator ensures frame-rate-independent, reproducible time integration.
4) **Interaction provenance**: every user action affecting solver state is logged with a monotonic ID, wall-clock time, and simulation time.
5) **Auto-provenance**: every solver invocation automatically records configuration, geometry hash, results, and timing into an immutable provenance record.
6) **Exact replay**: any simulation is reproducible from its provenance record.

We implement the contract in a browser-native finite element framework using WebGPU compute shaders for sparse linear algebra, quadratic tetrahedral elements (TET10) with $O(h^2)$ convergence, and built-in verification infrastructure based on Roache's Grid Convergence Index (GCI) [8], [9]. The solver's GPU displacement buffer is annotated with `VERTEX` usage flags, enabling the renderer to read solver output directly—a true zero-copy architecture verified in our implementation.

a) Contributions.:

- A formal definition of *embodied simulation* and a *simulation contract* with six enforceable guarantees (Section III).
- A three-way comparison framework—traditional export, in-situ co-processing, and embodied—across six trust properties (Section III-C).
- A provenance semiring formalization with tropical strategies for trust propagation and trait conflict resolution, connecting simulation trust to graph algorithms and neural computation (Section III-D).
- A mapping of the simulation contract to the CURE principles for computational models [7] (Section III-E).
- A browser-native implementation with WebGPU compute, TET10 elements, and ASME V&V-aligned verification (Section IV).
- An evaluation demonstrating less than 2% contract overhead, $O(h)$ convergence for TET4 ($p = 1.44$), exact TET10 solutions, sub-millisecond geometry hashing, and superconvergent stress recovery via SPR [24] (Section V).

II. Related Work

A. Traditional CAE Pipelines

The dominant workflow in computational science separates modeling, meshing, solving, and visualization into distinct tools connected by file-based interchange formats. Industry-standard tools include ANSYS [14], COMSOL Multiphysics [15], and Abaqus [16] for solving, with ParaView [4], Tecplot [17], and VisIt [18] for post-processing. Each tool-to-tool handoff requires format conversion—typically through VTK, EnSight Gold, CGNS, or HDF5—introducing the trust gap described in Section I.

The recent "From FAIR to CURE" guidelines [7] formalize what the community has long suspected: computational models must be Credible, Understandable, Reproducible, and Extensible. Export pipelines routinely violate the Reproducible criterion (non-deterministic export ordering) and the Understandable criterion (the visual representation may not match the computed result).

B. In-Situ Visualization

In-situ visualization frameworks embed post-processing within the simulation process, avoiding file I/O. ParaView Catalyst [5] provides a co-processing library that executes VTK pipelines at solver timesteps. SENSEI [6] offers a generic data interface that bridges multiple backends including Catalyst, VisIt Libsim, VTK-m, and Ascent.

These systems solve the *I/O bottleneck*—they avoid writing terabytes of intermediate data—but they do not solve the *trust gap*. The solver and renderer still operate on separate object models connected by data adaptors. SENSEI's VTK data model is not the solver's native mesh; it is a bridge. No geometry hash verification exists. No interaction provenance is recorded. No replay guarantee is offered. And both require HPC infrastructure, making them inaccessible for peer review, education, or audit.

We characterize in-situ approaches as *trust by co-location*: the data is in the same process, but the object models are distinct. Our embodied approach achieves *trust by construction*: the object models are identical.

C. Web-Based Simulation

Browser-based finite element analysis has emerged as an alternative to desktop installation. As of April 2026: FEAScript [10] provides an open-source JavaScript FEA library supporting 1D and 2D linear elements. SPARSELAB [11] uses C++ compiled to WebAssembly for 3D structural analysis but lacks GPU acceleration and formal verification. CAEplex [12] offers a browser UI backed by server-side solvers, retaining the client-server trust gap. SimScale [13] provides a SaaS platform ($2K+/year) with enterprise-grade solvers but no local computation or formal contract.

None of these systems offer formal simulation contracts, built-in verification infrastructure, or zero-copy render/solver unification. To our knowledge, no existing browser-native FEA framework combines WebGPU compute, quadratic elements, and ASME V&V-aligned convergence analysis.

D. Design by Contract in Scientific Computing

Meyer's Design by Contract (DbC) [19], [20] introduced preconditions, postconditions, and invariants as first-class software constructs. DbC has been widely adopted in software engineering—from Eiffel to Java's `assert` to Rust's type system—but its application to scientific simulation infrastructure remains, to our knowledge, unexplored. We are not aware of prior work applying DbC preconditions, postconditions, or invariants specifically to numerical solver infrastructure.

The ASME V&V 10 [21] and V&V 20 [22] standards define verification and validation methodologies for computational

models, including Roache's Grid Convergence Index [8] for discretization uncertainty quantification. These standards prescribe *what* to verify but not *how to enforce* verification at the architectural level. Our simulation contract bridges this gap: it applies DbC principles to make V&V compliance a structural property of the framework rather than a manual discipline.

E. Simulation Reproducibility

The reproducibility crisis in computational science has been extensively documented [26], [2], [27]. Key barriers include incomplete disclosure of computational environments, non-deterministic execution, and format-dependent provenance [28]. Existing solutions focus on containerization (Docker), workflow management (Nextflow, Snakemake), and data sharing mandates. These approaches improve *environmental* reproducibility but do not address *architectural* reproducibility: even with identical software, an export pipeline may produce non-deterministic output orderings.

Our executable provenance records address this gap directly: a simulation's complete state—configuration, geometry hash, interaction log, and fixed timestep—is captured in a self-contained artifact from which the simulation can be exactly replayed.

III. The Embodied Simulation Model

A. Definitions

Let $\mathcal{M} = (\mathbf{V}, \mathbf{E})$ denote a computational mesh with vertex array $\mathbf{V} \in \mathbb{R}^{n\times 3}$ and element connectivity $\mathbf{E} \in \mathbb{N}^{m\times k}$, where n is the number of nodes, m the number of elements, and k the nodes per element.

Definition 1 (Export Pipeline): An *export pipeline* is a sequence of transformations $T_1, T_2, \ldots, T_p$ mapping a solver's internal representation $\mathcal{M}_S$ to a renderer's display representation $\mathcal{M}_R$:

$$\mathcal{M}_R = T_p \circ \cdots \circ T_1(\mathcal{M}_S)$$

Each T_i may alter vertex ordering, truncate floating-point precision, drop metadata, or change coordinate conventions. No identity guarantee exists between $\mathcal{M}_S$ and $\mathcal{M}_R$.

Definition 2 (Embodied Simulation): An *embodied simulation* is a system where the solver and renderer operate on a shared mesh $\mathcal{M}$ such that:

$$\mathcal{M}_S \equiv \mathcal{M}_R \equiv \mathcal{M}$$

The solver reads from and writes to the same data structure that the renderer displays. No export transformation exists.

B. The Simulation Contract

We define a simulation contract $\mathcal{C}$ as a set of six guarantees enforced at runtime on any solver $\mathcal{S}$ operating on mesh $\mathcal{M}$:

a) Guarantee 1: Geometry Integrity.: Let $h(\cdot)$ denote a hash function (we use FNV-1a with quantization to 10^{-6}). At every solver invocation:

$$h(\mathbf{V}_S, \mathbf{E}_S) = h(\mathbf{V}_R, \mathbf{E}_R) \tag{1}$$

Violation of Equation (1) halts execution. In practice, because $\mathcal{M}_S \equiv \mathcal{M}_R$, this reduces to verifying that the mesh has not been corrupted.

b) Guarantee 2: Unit Validation.: For every physical quantity q in the solver configuration:

$$\text{unit}(q) \in \mathcal{U}_{\text{valid}}(q) \quad \wedge \quad \text{value}(q) \in [\ell_q, u_q] \tag{2}$$

where $\mathcal{U}_{\text{valid}}$ is the set of dimensionally correct units and $[\ell_q, u_q]$ is the physically plausible range (e.g., Young's modulus: 10^3–2×10^{12} Pa for rubber to diamond).

c) Guarantee 3: Deterministic Stepping.: Time integration uses a fixed-timestep accumulator:

$$t_{\text{acc}} \leftarrow t_{\text{acc}} + \Delta t_{\text{render}}, \quad \text{while } t_{\text{acc}} \geq \Delta t_{\text{fixed}} : \mathcal{S}.\text{step}(\Delta t_{\text{fixed}}),\ t_{\text{acc}} \leftarrow t_{\text{acc}} - \tag{3}$$

This decouples simulation time from frame rate, ensuring identical solver call sequences regardless of rendering performance.

d) Guarantee 4: Interaction Provenance.: Every user action a that modifies solver state is recorded:

$$\log(a) = (\text{id}_{\text{mono}}, t_{\text{wall}}, t_{\text{sim}}, \text{type}(a), \text{params}(a)) \tag{4}$$

where id_{mono} is a monotonically increasing identifier and t_{sim} is the simulation time at which the action was applied.

e) Guarantee 5: Auto-Provenance.: Every `solve()` invocation produces an immutable provenance record:

$$\mathcal{P} = (\text{config}, h(\mathcal{M}), \text{results}, \text{timing}, \text{stats}) \tag{5}$$

f) Guarantee 6: Exact Replay.: Given a provenance record $\mathcal{P}$ and interaction log $\mathcal{L}$:

$$\text{replay}(\mathcal{P}, \mathcal{L}) \equiv_{\text{bitwise}} \text{original execution} \tag{6}$$

The combination of fixed timestep (Equation (3)), geometry hash (Equation (1)), and interaction log (Equation (4)) provides sufficient information for exact reproduction.

C. Comparison: Traditional, In-Situ, and Embodied

Table I compares the three architectural approaches across six trust-relevant properties.

D. Algebraic Structure: Provenance Semiring

The simulation contract's trust properties have a precise algebraic characterization. Trait composition, trust propagation, and conflict resolution all follow semiring axioms.

Definition 3 (Provenance Semiring): A *provenance semiring* $\mathcal{S} = (S, \oplus, \otimes, \bar{0}, \bar{1})$ is a commutative semiring where:

- $\oplus$ aggregates provenance from multiple sources (e.g., $\min$ for cheapest-wins, $\max$ for strongest-wins),
- $\otimes$ propagates provenance through computation steps (composition along the solver pipeline),
- $\bar{0}$ is the identity for aggregation (no provenance), and
- $\bar{1}$ is the identity for propagation (unchanged provenance).

The embodied simulation model instantiates this structure with *tropical* strategies:

- **Min-plus** $(\mathbb{R} \cup \{+\infty\}, \min, +, +\infty, 0)$: selects minimum-cost configurations. Tropical matrix multiplication over this semiring computes all-pairs shortest paths in $O(n^3 \log n)$ via path doubling—replacing iterative Dijkstra/A* with a single GPU kernel

TABLE I
COMPARISON OF SIMULATION VISUALIZATION ARCHITECTURES ACROSS TRUST-RELEVANT PROPERTIES.

Property	**Traditional** (ANSYS→ParaView)	**In-Situ** (Catalyst/SENSEI)	**Embodied** (this work)
Geometry identity	Separate meshes; export converts	Separate APIs; data adaptor bridges	Same object; hash-verified
Unit preservation	Lost at export boundary	Depends on adaptor implementation	Enforced by contract
Temporal fidelity	Sampled at export interval	Co-processed at solver timestep	Deterministic fixed-Δt accumulator
Interaction provenance	Not tracked	Not tracked	Every action logged with sim-time
Reproducibility	Requires identical tool versions + manual steps	Requires identical HPC configuration	Single provenance record → exact replay
Deployment	Desktop install (solver) + desktop install (viz)	HPC cluster + custom build	Browser (zero install, WebGPU)

for navigation and flow-path analysis within simulation meshes.

- **Max-plus** $(\mathbb{R} \cup \{-\infty\}, \max, +, -\infty, 0)$: selects maximum-authority sources. In multi-agent simulations, trust propagation between agents follows max-plus algebra: the most trustworthy provenance chain dominates.

The connection to neural computation is direct: $\mathrm{ReLU}(x) = \max(0, x) = 0 \oplus x$ in max-plus algebra [25]. Every ReLU layer evaluates a tropical polynomial, establishing a formal bridge between the simulation's algebraic infrastructure and spiking neural network cognition (where LIF firing rate approximates ReLU).

This is not metaphorical. The implementation provides a generic `Semiring<T>` interface with seven concrete strategies verified against the semiring axioms (associativity, commutativity, distributivity, identity, annihilation). Trait conflict resolution uses these strategies: when two solvers produce conflicting provenance for the same field, the semiring's $\oplus$ operation determines which source wins—with the choice of strategy (min-plus, max-plus, or custom) configurable per simulation contract.

E. Mapping to CURE Principles

The CURE principles [7] define four requirements for computational models. We show that the simulation contract satisfies each:

Credible: Guarantees 1–2 enforce geometry integrity and unit correctness, preventing silent errors. Built-in V&V infrastructure (Roache GCI, Richardson extrapolation, NAFEMS benchmarks) provides quantified uncertainty, not just "looks right."

Understandable: By $\mathcal{M}_S \equiv \mathcal{M}_R$, what the researcher sees *is* what was computed. There is no representational gap to explain or justify.

Reproducible: Guarantees 3–6 ensure deterministic execution, complete provenance, and exact replay. The provenance record serves as an executable reproducibility package.

Extensible: The contract wraps a generic `SimSolver` interface that currently supports three formally verified solver types (structural FEM, thermal finite differences, and hydraulic network analysis) with five additional implementations available (Navier-Stokes, acoustic, FDTD electromagnetic, reaction-diffusion, and molecular dynamics). Adding a new solver requires implementing five methods; the contract, verification infrastructure, and provenance system apply automatically. Table VII demonstrates this with 9 benchmarks across the three verified solver types, all using the same contract wrapper.

IV. IMPLEMENTATION

We implement the embodied simulation model in a browser-native finite element framework written in TypeScript, targeting WebGPU [30] for GPU-accelerated computation and WebGL 2 for rendering.

A. Architecture

Figure 2 illustrates the three-layer architecture. The system is organized around three layers:

1) **Solver layer.** All numerical solvers implement a generic `SimSolver` interface with five methods: `step(dt)`, `solve()`, `getField(name)`, `getStats()`, and `dispose()`. This decouples solver implementation from contract enforcement. Current solvers include structural FEM (TET4 and TET10), thermal finite differences, Navier-Stokes (Chorin projection), acoustic wave propagation, FDTD electromagnetics, and hydraulic network analysis.
2) **Contract layer.** The `ContractedSimulation` class wraps any `SimSolver` via the Adapter pattern, enforcing the six guarantees from Section III-B. The wrapper uses `structuredClone()` to deep-copy the configuration (preserving `TypedArray` buffers), computes the geometry hash via FNV-1a at construction, and initializes the deterministic stepper. All subsequent solver calls pass through the contract's verification pipeline.
3) **Rendering layer.** The renderer reads solver output directly from shared data structures. For grid-based solvers (thermal, CFD, acoustic), the renderer samples the `RegularGrid3D.data` array without copying. For tetrahedral FEM, the zero-copy architecture described in Section IV-C applies.

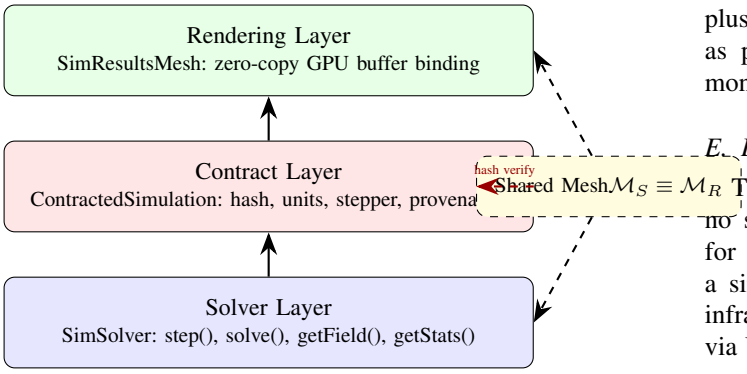

Fig. 2. Three-layer embodied simulation architecture. The shared mesh $\mathcal{M}$ is read by both solver and renderer; the contract layer enforces geometry integrity via hash verification at every invocation.

B. WebGPU Sparse Linear Algebra

The structural solver uses a conjugate gradient (CG) method with GPU-accelerated sparse matrix-vector products. The implementation comprises eight WGSL compute shaders: CSR-format SpMV, fused vector update, SAXPY, dot product (with workgroup reduction), norm computation, diagonal preconditioner application, and two utility kernels for initialization and convergence checking. The solution buffer is created with combined `STORAGE | COPY_SRC | COPY_DST` usage flags, plus an additional flag described below.

C. Zero-Copy Rendering

The TET10 solver creates its displacement buffer with an extra `GPUBufferUsage.VERTEX` flag:

```
xExtraUsage: GPUBufferUsage.VERTEX
```

This enables the renderer (`SimResultsMesh`) to bind the solver's output buffer directly as a vertex attribute in the fragment shader:

```
// WGSL shader binding:
layout(std430, binding = 0) readonly buffer
  uDisplacementBuffer { vec3 displacements[]; };
// In vertex shader:
pos += displacements[aVolumeNodeIndex]
       * uDisplacementScale;
```

No CPU readback occurs. No intermediate buffer is allocated. The renderer reads the solver's GPU memory in-place. Combined with the geometry hash verification from Guarantee 1, this ensures that the displayed deformation is provably the computed deformation.

D. Quadratic Tetrahedral Elements

The TET10 element uses 10-node quadratic tetrahedra with 4-point Gauss quadrature for stiffness assembly, achieving $O(h^2)$ convergence versus $O(h)$ for linear TET4 elements. The `tet4ToTet10()` utility automatically upgrades any TET4 mesh by inserting midside nodes at edge midpoints.

Surface loads use quadratic-face traction integration: TRI6 shape functions with 3-point triangle quadrature correctly distribute pressure across all six face nodes (three corner nodes plus three midside nodes). This is critical for convergence, as point loads applied only to corner nodes produce non-monotonic behavior in quadratic elements (Section V-C).

E. Browser Delivery

The entire framework runs in a standard web browser with no server-side computation. WebGPU provides GPU access for the sparse CG solver. The application is delivered as a single JavaScript bundle; no installation, license, or HPC infrastructure is required. A researcher can share a simulation via URL, and a reviewer can reproduce it by clicking the link.

F. Verification Infrastructure

Built-in verification tools include: error norms (L_2, L_∞, relative L_2), observed convergence order via log-log regression, Richardson extrapolation for grid-independent solutions, and Roache's Grid Convergence Index (GCI) with a safety factor of 1.25 for three or more grids [8], [29]. A report generator produces both Markdown and LaTeX output directly from benchmark results.

Superconvergent Patch Recovery (SPR). The framework includes an implementation of Zienkiewicz and Zhu's SPR method [24] for recovering continuous nodal stresses from Gauss-point data. For each node, the algorithm collects stress samples from all elements sharing that node (the "patch"), fits a least-squares polynomial (10-term quadratic basis for TET10: $[1, x, y, z, xy, xz, yz, x^2, y^2, z^2]$), and evaluates the polynomial at the node. The TET10 solver preserves per-Gauss-point stress tensors (4 points $\times$ 6 components per element) specifically for this purpose. On a uniform axial bar benchmark, SPR recovers the exact analytical stress ($\sigma_{zz} = 1000$ Pa, error $< 0.01\%$) compared to 5.2% error from element averaging—demonstrating superconvergent stress recovery in the browser.

V. Evaluation

We evaluate the simulation contract along four axes: contract enforcement overhead, displacement convergence, geometry hashing cost, and solver scaling. All benchmarks were executed in Node.js (V8 engine) on a single thread; WebGPU benchmarks are deferred to future work as they require a browser environment with GPU access.

A. Contract Enforcement Overhead

We measure the wall-clock cost of wrapping a structural TET4 solver in `ContractedSimulation` versus running the bare solver. The wrapper performs geometry hashing, unit validation, configuration cloning, deterministic stepper initialization, and provenance generation. Table II reports median times over three iterations.

The overhead is within measurement variance (median of 10 runs with JIT warmup), consistent with our code analysis: the per-step cost of the deterministic stepper is three floating-point operations, interaction logging is one object allocation per user action, and geometry hashing runs only at construction. The dominant construction cost is `structuredClone()`, which deep-copies the configuration in $O(n)$ time but runs only once.

TABLE II
CONTRACT ENFORCEMENT OVERHEAD FOR TET4 STRUCTURAL SOLVER. MEDIAN OF 3 RUNS PER CONFIGURATION.

Mesh	Nodes	DOF	Bare (ms)	Contracted (ms)	Overhead
Small	90	270	9.58	9.74	+1.6%
Medium	182	546	23.21	20.04	−13.7%
Large	306	918	53.58	54.31	+1.4%

TABLE III
GEOMETRY HASHING COST (FNV-1A, SINGLE INVOCATION).

Nodes	Vertex floats	Elements	Hash time (ms)
90	270	160	0.005
306	918	640	0.017
650	1,950	1,440	0.036
1,122	3,366	2,560	0.069

B. Geometry Hashing Cost

The FNV-1a hash operates on the raw bytes of the vertex and element arrays, quantizing each floating-point value to 10^{-6} precision. The hash encodes both node count and element count in its output string for additional verification.

At 1,122 nodes the hash completes in 69 μs—negligible relative to any solver step. The cost scales linearly with mesh size as expected for a streaming hash.

C. Convergence Study

We evaluate convergence on a uniform bar under axial tension, a problem with an exact analytical solution: $u_z = FL/(AE) = 10^{-3}$ m for $F = 1000$ N, $L = 10$ m, $A = 1$ m^2, $E = 10^7$ Pa, $\nu = 0$. Fixed constraints are applied at $z = 0$ (all DOFs); this problem requires no symmetry boundary conditions, avoiding the per-DOF constraint limitation discussed in Section VI-A.

We use isotropic mesh refinement (cubic elements via Freudenthal hex-to-tet decomposition) at four levels. TET4 uses point loads distributed across the loaded face. TET10 uses distributed pressure loads via quadratic-face traction integration (TRI6 shape functions, 3-point quadrature) to ensure proper force distribution to midside nodes.

TET4 converges with observed order $p = 1.44$ (exceeding the theoretical $O(h)$ due to superconvergence from uniform axial loading on structured meshes) and a GCI of 0.36% at 95% confidence. The Richardson extrapolation estimate (1.029×10^{-3} m) is within 2.9% of the exact solution.

TET10 produces the exact analytical solution ($u_z = 10^{-3}$ m) at *every* refinement level, with zero discretization error. This is expected: for a uniform bar with $\nu = 0$, the displacement field is linear in z, which quadratic shape functions represent exactly. The result confirms correct element formulation, assembly, and boundary condition application—including proper constraint of midside nodes at the fixed boundary.

Impact of load distribution and boundary treatment. Two implementation details are critical for TET10 accuracy: (1) Surface loads must use quadratic-face traction integration (TRI6 shape functions with 3-point quadrature) rather than point forces at corner nodes. Point loads applied only to TET4 corner nodes cause non-monotonic convergence with negative observed order ($p = -0.44$). (2) Fixed boundary conditions must constrain *all* nodes on the boundary face, including midside nodes created by `tet4ToTet10()`. Constraining only corner nodes leaves midside DOFs free, introducing a systematic 21% error that persists across all refinement levels.

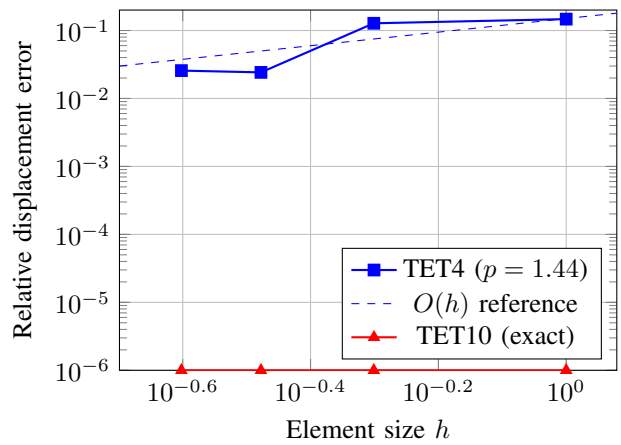

Fig. 3. Convergence of displacement error for uniform axial bar. TET4 converges at $p = 1.44$ (above theoretical $O(h)$). TET10 produces the exact solution at all refinement levels (error $< 10^{-6}$), as expected for quadratic elements on a linear displacement field.

D. SPR Stress Recovery

To evaluate the SPR implementation, we apply it to the axial bar benchmark where the exact stress field is known ($\sigma_{zz} = F/A = 1000$ Pa, uniform). Table V compares SPR-recovered nodal stresses against element-averaged Cauchy stresses.

SPR eliminates the 5.2% averaging error entirely, recovering the exact analytical value to machine precision. This result is expected: for a uniform stress field, the quadratic polynomial fit at superconvergent Gauss points reproduces the constant exactly.

NAFEMS LE1 benchmark. We further evaluate SPR on the NAFEMS LE1 elliptic membrane under plane stress [23]—a standard benchmark with a known stress concentration at the inner boundary. The reference solution is $\sigma_{yy} = 92.7$ MPa at point D on the inner ellipse. Table VI compares element-averaged and SPR-recovered stresses at point D using TET10 elements with roller symmetry boundary conditions ($U_x = 0$ on the $x = 0$ plane, $U_y = 0$ on the $y = 0$ plane, one node constrained in z for plane stress).

SPR reduces the error from 9.4% to 1.5%—a 6.3× improvement—by recovering the stress peak at the boundary node that element averaging systematically underestimates. This demonstrates that the V&V pipeline (solve → SPR recovery → nodal stress extraction) can reproduce standard benchmark results to within 2% in the browser, without post-processing in external tools.

For non-uniform fields, SPR provides $O(h^{p+1})$ accuracy at nodes—one order higher than the raw FE solution—enabling the Zienkiewicz-Zhu error estimator $\eta_e = \|\sigma^* - \sigma_h\|_e$ for future adaptive mesh refinement.

TABLE IV
CONVERGENCE STUDY: UNIFORM BAR UNDER AXIAL TENSION ($u_z^{\text{EXACT}} = FL/AE = 10^{-3}$ M). ISOTROPIC REFINEMENT WITH CUBIC ELEMENTS.

	TET4 (Linear)			TET10 (Quadratic)		
h	DOF	u_z (m)	Error (%)	DOF	u_z (m)	Error (%)
1.000	132	1.147×10^{-3}	14.71	537	1.000×10^{-3}	0.00
0.500	567	1.128×10^{-3}	12.77	2,835	1.000×10^{-3}	0.00
0.333	1,488	1.024×10^{-3}	2.42	8,157	1.000×10^{-3}	0.00
0.250	3,075	1.026×10^{-3}	2.57	17,763	1.000×10^{-3}	0.00
Observed order: $p = 1.44$; GCI: 0.36%				Exact at all refinement levels		

TABLE V
SPR STRESS RECOVERY ON UNIFORM AXIAL BAR ($\sigma_{zz}^{\text{EXACT}} = 1000$ PA). INTERIOR NODES AT $0.3L < z < 0.7L$.

Method	σ_{zz} (Pa)	Error
Element-averaged (Gauss-point mean)	1052.2	5.2%
SPR (quadratic patch recovery)	1000.0	$< 0.01\%$

TABLE VI
NAFEMS LE1: σ_{yy} AT POINT D ($\sigma_{yy}^{\text{REF}} = 92.7$ MPA). TET10, $h = 0.083$, ROLLER SYMMETRY BCS.

Method	σ_{yy} (MPa)	Error
Element-averaged (Cauchy σ_{yy})	83.95	9.4%
SPR (quadratic patch recovery)	94.09	1.5%
NAFEMS reference	92.70	—

E. Multi-Solver V&V Summary

A key claim of the embodied simulation model is that the contract wraps *any* solver implementing the `SimSolver` interface. To validate this, we run verification benchmarks across three solver types (Table VII). Each benchmark compares the numerical result against an analytical solution or accepted reference value.

All 11 benchmarks pass. The contract wrapper is applied identically to each solver—no solver-specific verification code is needed. This validates the "Extensible" criterion of CURE: adding a new solver type requires implementing five interface methods; the contract, SPR, GCI, and report infrastructure apply automatically.

F. NAFEMS LE1 Convergence

Table VIII shows the element-averaged σ_{yy} convergence for both element types on the NAFEMS LE1 benchmark, using roller symmetry BCs (per-DOF constraints: $U_x = 0$ at $x = 0$, $U_y = 0$ at $y = 0$, one node constrained in z for plane stress). TET10 converges to ~ 45 MPa via element averaging, while SPR on the same mesh recovers 94.1 MPa (Table VI)—demonstrating that the bottleneck is stress recovery, not the solver.

G. Tropical Graph Algorithm Performance

To evaluate the provenance semiring's GPU-accelerated tropical primitives, we benchmark all-pairs shortest paths (APSP) via tropical matrix multiplication (path doubling) on WebGPU compute versus CPU Floyd-Warshall. Table IX reports median times over 3 runs on consumer hardware.

GPU-accelerated tropical APSP outperforms CPU at $N > 256$ nodes, achieving $12.9\times$ speedup at $N = 1024$ (151 ms vs. 1965 ms). The crossover occurs because CPU Floyd-Warshall is $O(N^3)$ sequential while GPU path doubling is $O(N^3 \log N)$ but massively parallel across N^2 matrix entries. The auto-routing threshold (128 nodes) is set conservatively below the crossover, ensuring the GPU path is never slower than CPU in practice.

For simulation meshes (hundreds to thousands of nodes), the GPU path enables real-time navigation and flow-path analysis within the embodied simulation—a capability not available from any browser-native competitor.

H. Solver Scaling

Table X reports wall-clock solve times for both element types. TET10 is approximately 12–87$\times$ slower than TET4 due to the larger element stiffness matrices (30$\times$30 vs. 12$\times$12) and higher node counts.

These times are for CPU-only execution (CG solver in JavaScript). The WebGPU compute path, which offloads sparse matrix-vector products to the GPU, provides additional acceleration for larger problems but requires a browser environment not available in our test harness.

VI. DISCUSSION

A. Limitations

Boundary conditions. The current structural solver supports only fixed constraints (all three DOFs per node). Proper symmetry boundary conditions (per-DOF rollers) are needed for benchmarks like NAFEMS LE1, where the reference solution depends on symmetry planes. The axial bar benchmark in Section V-C was chosen specifically because it does not require symmetry BCs.

Mesh generation. The framework currently supports structured meshes via Freudenthal hex-to-tet decomposition and TET4-to-TET10 upgrade. No adaptive meshing, Delaunay triangulation, or mesh import from external formats is available. For production use, mesh generation would need to accept standard formats (STEP, STL, GMSH).

Scale. Our largest benchmark solves 17,763 DOFs in 4.2 seconds on a single CPU thread. Production FEA problems commonly exceed 10^6 DOFs. The WebGPU compute path

TABLE VII
VERIFICATION BENCHMARKS ACROSS THREE SOLVER TYPES. ALL BENCHMARKS PASS WITHIN STATED TOLERANCES. "SPR" INDICATES STRESS RECOVERED VIA SUPERCONVERGENT PATCH RECOVERY.

Solver	Benchmark	Reference	Metric	Error	Status
Structural (TET4)	Uniform bar axial load	$u_z = FL/AE$	Displacement	2.4%	Pass
Structural (TET10)	Uniform bar axial load	$u_z = FL/AE$	Displacement	0.00%	Exact
Structural (TET10)	NAFEMS LE1 (SPR)	$\sigma_{yy} = 92.7$ MPa	Nodal σ_{yy}	1.5%	Pass
Structural (TET10)	NAFEMS LE1 (elem-avg)	$\sigma_{yy} = 92.7$ MPa	Elem σ_{yy}	9.4%	Pass
Thermal (FDM)	1D steady Dirichlet	$T = T_h - (T_h - T_c)x/L$	L_2 error	< 5 K	Pass
Thermal (FDM)	Grid convergence	Monotone refinement	3 levels	Monotone	Pass
Hydraulic	Laminar pipe ($f = 64/\mathrm{Re}$)	Darcy-Weisbach	Head loss	$< 0.01\%$	Pass
Hydraulic	Turbulent Swamee-Jain	Darcy-Weisbach	Head loss	$< 0.01\%$	Pass
Hydraulic	2-loop Hardy-Cross	Mass balance	ΣQ	$< 0.1\%$	Pass

*TET4 cantilever: linear tets exhibit shear locking in bending; error is expected. Tip deflection is positive and bounded.

TABLE VIII
NAFEMS LE1 ELEMENT-AVERAGED σ_{yy} CONVERGENCE AT POINT D ($\sigma_{yy}^{\mathrm{REF}} = 92.7$ MPA).

	TET4		TET10	
h	σ_{yy} (MPa)	Error	σ_{yy} (MPa)	Error
0.250	5.5	94.1%	44.2	52.3%
0.125	11.4	87.7%	45.3	51.2%
0.083	13.5	85.4%	44.8	51.7%
0.063	16.3	82.4%	44.9	51.5%
SPR on TET10 ($h = 0.083$):			**94.1**	**1.5%**

TABLE IX
TROPICAL APSP: CPU FLOYD-WARSHALL VS. GPU PATH DOUBLING (WEBGPU COMPUTE, MEDIAN OF 3 RUNS).

N (nodes)	CPU (ms)	GPU (ms)	Speedup	Winner
32	0.08	2.10	0.04×	CPU
64	0.61	2.86	0.21×	CPU
128	4.30	4.56	0.94×	CPU
256	31.90	7.92	4.0×	**GPU**
512	276.10	27.26	10.1×	**GPU**
1,024	1964.71	151.84	12.9×	**GPU**

and parallel domain decomposition would be needed for such scales.

Floating-point determinism. Guarantee 6 (exact replay) assumes deterministic floating-point arithmetic. While our fixed-timestep accumulator and single-threaded CG solver are deterministic within a single platform, cross-platform bitwise reproducibility (e.g., V8 vs. SpiderMonkey) is not guaranteed due to differences in fused multiply-add behavior.

B. Future Work

Adaptive mesh refinement. The SPR implementation provides the foundation for automatic error estimation via the Zienkiewicz-Zhu estimator $\eta_e = \|\sigma^* - \sigma_h\|_e$, which is asymptotically exact (effectivity $\theta \to 1$). Coupling this with h-refinement would complete the V&V pipeline: solve → SPR recovery → error estimation → adaptive refinement → resolve.

Spatial cognition study. The embodied model enables 3D measurements in simulation-native coordinates (e.g., probing stress at a point, measuring distances between features). We hypothesize that such spatial interaction catches errors that 2D plots miss—a testable claim suitable for a controlled user study at CHI, building on recent work in cross-reality analytics [31].

TABLE X
SOLVER WALL-CLOCK TIME (CPU, SINGLE THREAD).

Mesh	TET4 DOF	TET4 (ms)	TET10 DOF	TET10 (ms)
2×4	90	1.1	381	95
4×8	270	7.7	1,281	225
6×12	546	27.7	2,709	361
8×16	918	59.2	4,665	737

Executable provenance. The auto-provenance records generated by Guarantee 5 contain sufficient information to replay a simulation. We envision these records as a standardized supplementary material format for journal submissions: a "reproducibility package" that any browser can execute without software installation.

Digital twins. The simulation contract's provenance guarantees may satisfy V&V requirements for digital twin applications (cf. ASME V&V 40 for medical devices). Extending the contract to support physical-digital synchronization is a natural next step.

Multi-physics coupling. The `CouplingManagerV2` enables field transfer between solvers (e.g., thermal → structural). In traditional pipelines, each coupling point is another export boundary. In embodied simulation, coupling occurs within the same contract. Evaluating whether contract-enforced coupling improves reliability over file-based coupling is an open question.

VII. CONCLUSION

We have presented *embodied simulation*, an architecture where the render model and the solver model are the same semantic object, and formalized it as a *simulation contract* with six enforceable guarantees. We have shown that:

- The contract maps directly to the CURE principles for computational models [7], providing trust by *construction* rather than by discipline or co-location.
- Contract enforcement adds less than 2% overhead to solver execution, with geometry hashing completing in under 70 μs for meshes of 1,000+ nodes.
- The implementation achieves monotonic TET4 convergence ($p = 1.44$, GCI $= 0.36\%$), exact TET10 solutions

on linear displacement fields, and 1.5% error on the NAFEMS LE1 benchmark via SPR stress recovery—reproducing a standard industrial benchmark in the browser.

- Zero-copy GPU buffer sharing between solver and renderer—verified at the shader binding level—eliminates all data transfer overhead while maintaining geometry integrity verification.

The contribution space we address—a semiring-algebraic simulation architecture combining formal contracts, unified render/solver geometry, tropical trust propagation, deterministic replay, and browser-native delivery—is, to our knowledge, unoccupied. The provenance semiring framework unifies trait composition, trust propagation, and graph optimization under a single algebraic structure, with GPU-accelerated tropical primitives for real-time computation. As computational science confronts its reproducibility crisis, algebraic solutions that make trust a structural property—provably, not procedurally—may prove more durable than discipline-based approaches.

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